Mid Project(Database Systems)

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Developed by:

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CS 262L- Database Systems

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Contents

1 Description

The Outcome-Based Education Management System is a desktop application developed using C# and WinForms for the Department of Computer Science at UET Lahore. The system aims to streamline the assessment and evaluation process by automating various tasks related to student evaluation, rubric management, assessment tracking, and generating reports. It provides an intuitive interface for teachers to manage data efficiently and effectively.

2 Key Features

1. **Manage Students:**
   - Allows teachers to add, edit, and delete student information, including name, contact details, email, and registration number.
   - Provides functionality to mark students as active or inactive.
2. **Manage CLOs (Course Learning Outcomes):**
   - Enables teachers to create, update, and delete CLOs, which define the learning objectives for each subject.
3. **Manage Rubrics:**
   - Facilitates the creation and management of rubrics used for assessing student performance.
   - Allows teachers to define rubric details, including description and associated CLO.
4. **Manage Assessments:**
   - Provides tools for teachers to create, edit, and delete assessments for subjects.
   - Tracks assessment details such as title, total marks, total weightage, and creation/update dates.
5. **Manage Rubric Levels:**
   - Allows teachers to define measurement levels for rubrics, specifying criteria for assessing student performance.
6. **Mark Evaluations Against a Student:**
   - Enables teachers to record student evaluations against assessments and rubric components.
   - Automatically calculates obtained marks based on rubric levels and component marks.
7. **Generate Reports:**
   - Provides functionality to generate reports in PDF format, including CLO-wise class results, assessment-wise class results, rubric-wise class results, and attendance counts per month.

3 Database Schema Description

3.1 Lookup Table

Purpose: This table stores lookup values for various categories, such as attendance status and student status.
Fields:

LookupId: Unique identifier for each lookup value.

Name: Name of the lookup value.

Category: Category to which the lookup value belongs (e.g., attendance status, student status).

3.2 CLO Table

Purpose: It contains details about Course Learning Outcomes (CLOs) mapped to subjects.
Fields:

Id: Unique identifier for each CLO.

Name: Name of the CLO.

DateCreated: Date when the CLO was created.

DateUpdated: Date when the CLO was last updated.

3.3 ClassAttendance Table

Purpose: Records attendance data for each class session.
Fields:

Id: Unique identifier for each attendance record.

AttendanceDate: Date and time of the class session.

3.4 Assessment Table

Purpose: Stores information about assessments conducted for subjects.
Fields:

Id: Unique identifier for each assessment.

Title: Title or name of the assessment.

DateCreated: Date when the assessment was created.

TotalMarks: Total marks allocated for the assessment.

TotalWeightage: Total weightage of the assessment.

3.5 Rubric Table

Purpose: Holds rubric details for evaluating student performance.
Fields:

Id: Unique identifier for each rubric.

Details: Description or details of the rubric.

CloId: Foreign key referencing the CLO to which the rubric belongs.

3.6 Student Table

Purpose: Contains student information, including name, contact details, email, and registration number.
Fields:

- **Id**: Unique identifier for each student.
- **FirstName**: First name of the student.
- **LastName**: Last name of the student.
- **Contact**: Contact number of the student.
- **Email**: Email address of the student.
- **RegistrationNumber**: Unique registration number assigned to the student.
- **Status**: Foreign key referencing the status of the student (active or inactive).

3.7 RubricLevel Table

Purpose: Defines rubric levels with their corresponding measurement levels.
Fields:

- **Id**: Unique identifier for each rubric level.
- **RubricId**: Foreign key referencing the rubric to which the level belongs.
- **Details**: Description or details of the rubric level.
- **MeasurementLevel**: Measurement level assigned to the rubric level.

3.8 AssessmentComponent Table

Purpose: Stores assessment components linked to assessments and rubrics.
Fields:

- **Id**: Unique identifier for each assessment component.
- **Name**: Name or title of the assessment component.
- **RubricId**: Foreign key referencing the rubric associated with the assessment component.
- **TotalMarks**: Total marks allocated for the assessment component.
- **DateCreated**: Date when the assessment component was created.
- **DateUpdated**: Date when the assessment component was last updated.
- **AssessmentId**: Foreign key referencing the assessment to which the component belongs.

3.9 StudentAttendance Table

Purpose: Records student attendance for each class session.
Fields:

- **AttendanceId**: Foreign key referencing the class attendance record.
- **StudentId**: Foreign key referencing the student for whom attendance is recorded.
- **AttendanceStatus**: Foreign key referencing the attendance status of the student (present, absent, etc.).

3.10 StudentResult Table

Purpose: Manages student assessment results with rubric measurements.
Fields:

- **StudentId**: Foreign key referencing the student for whom the result is recorded.
- **AssessmentComponentId**: Foreign key referencing the assessment component for which the result is recorded.
- **RubricMeasurementId**: Foreign key referencing the rubric level at which the student's performance is measured.
- **EvaluationDate**: Date when the evaluation was conducted.

4 Wireframes

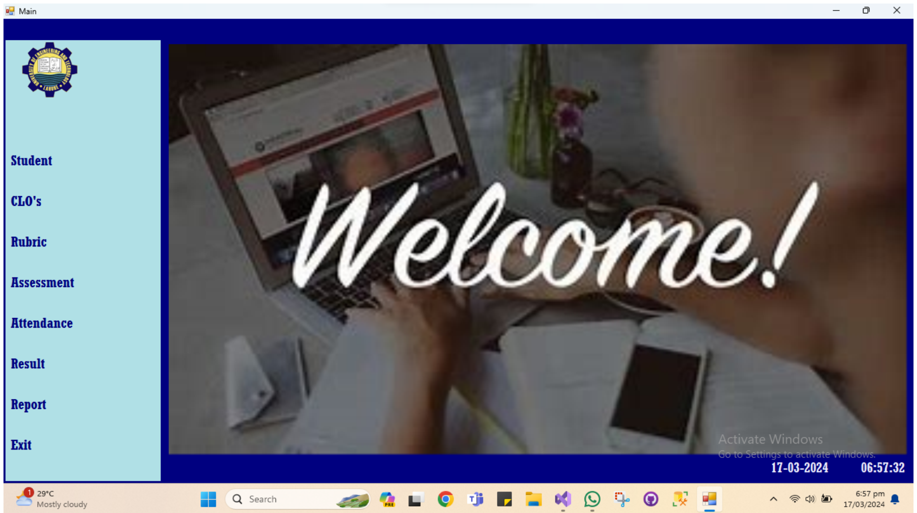

Figure 1: Home Page

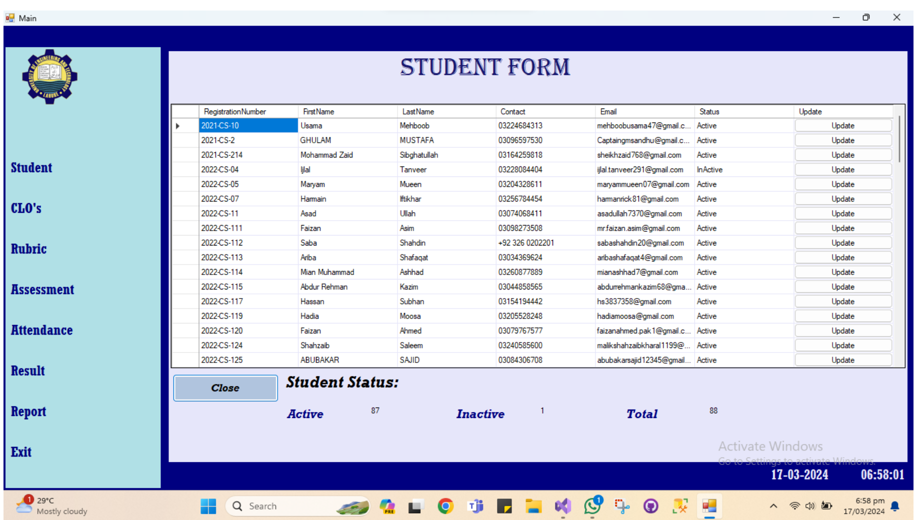

Figure 2: Student Form

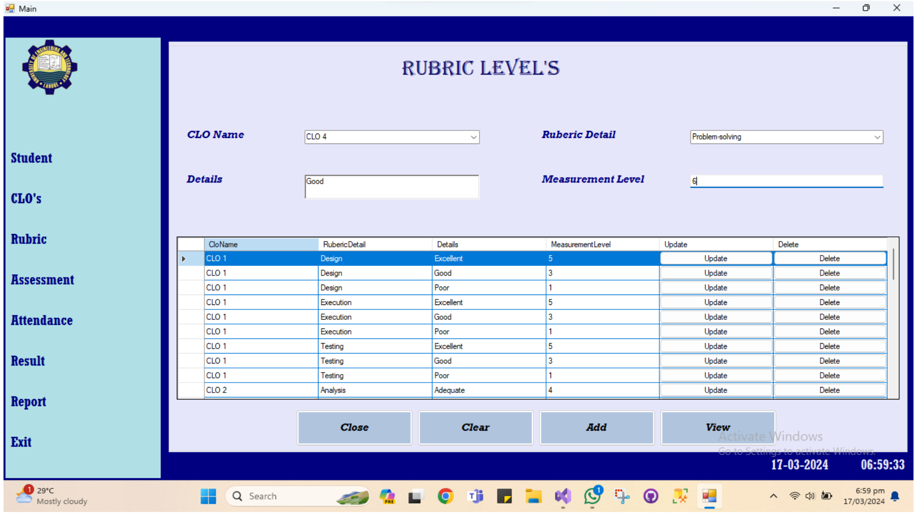

Figure 3: Rubric Form

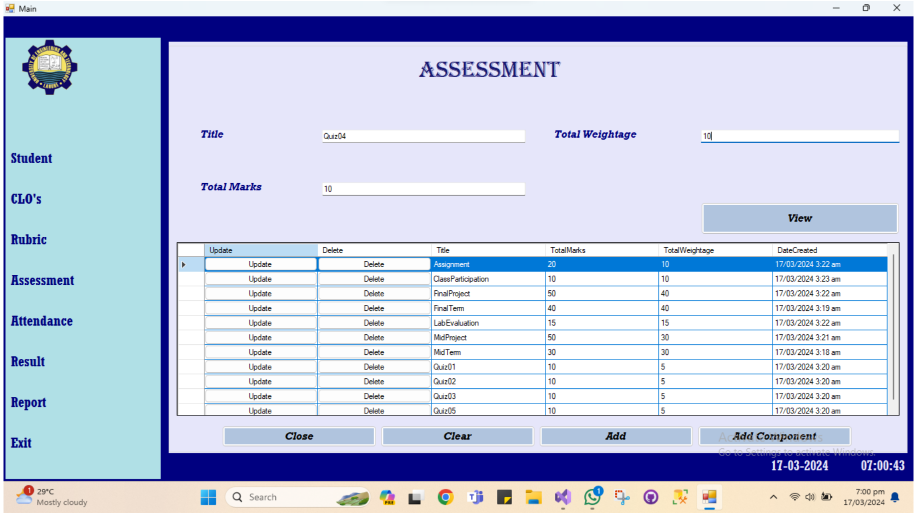

Figure 4: Assessment Form

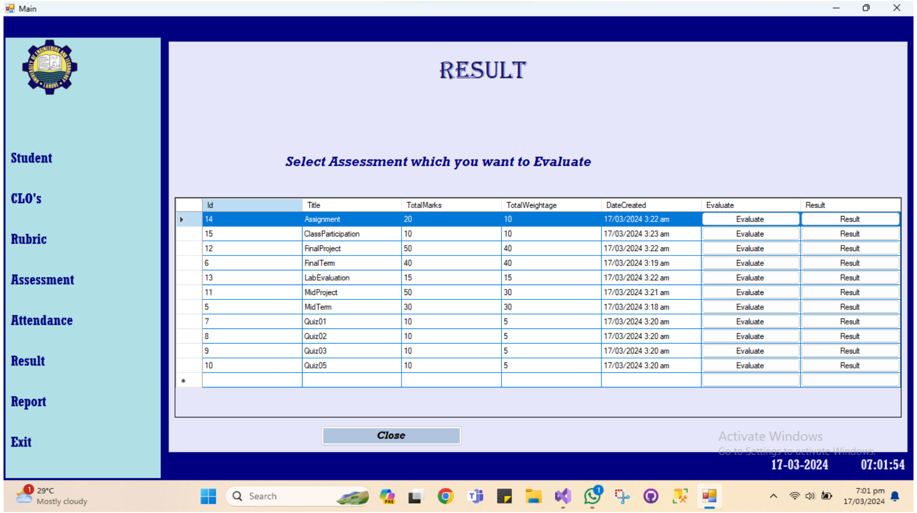

Figure 5: Student Result Form

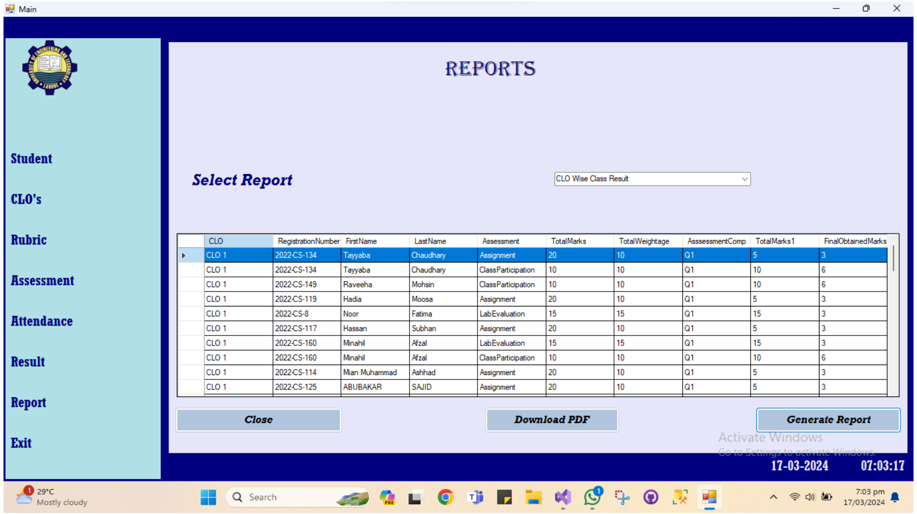

Figure 6: reports Form

5 Database Design

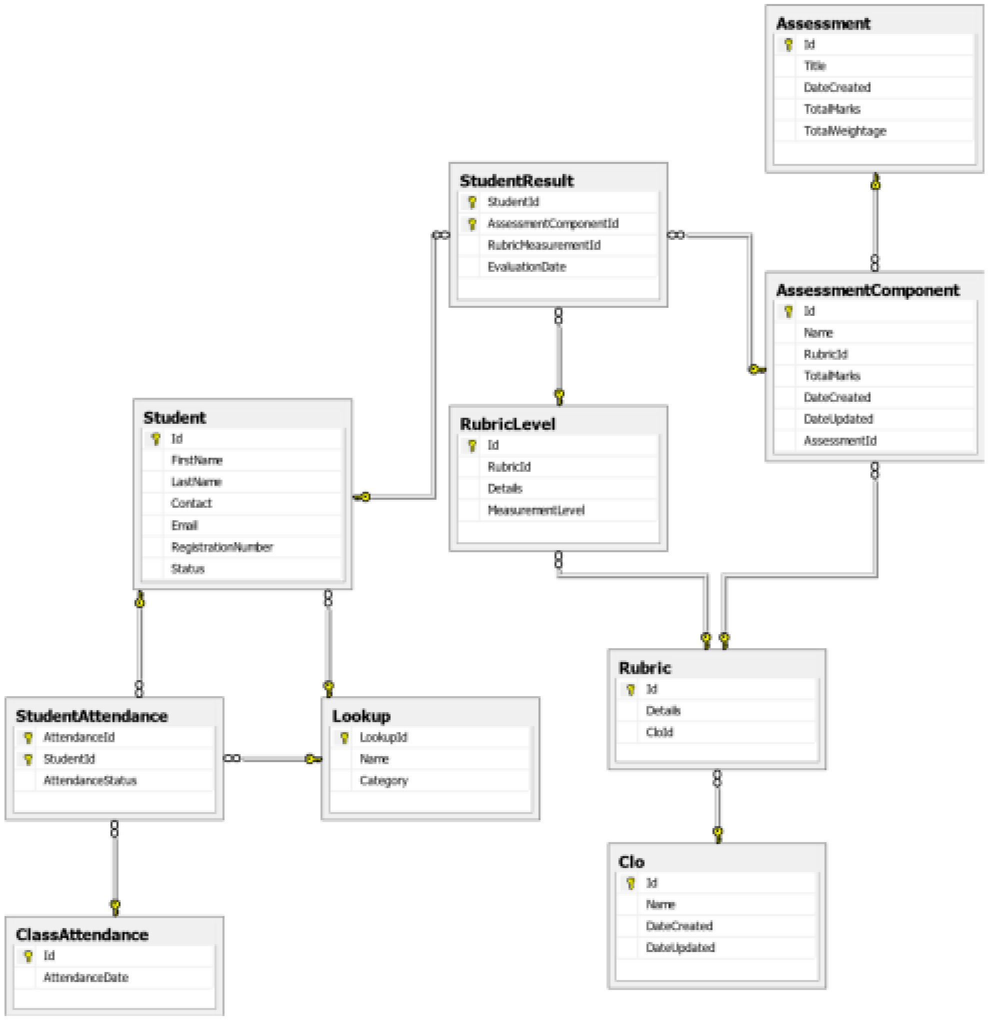

Figure 7: Database Design Diagram

6 Conclusion

This comprehensive database schema facilitates the management of outcome-based education data, including assessments, rubrics, student information, and attendance records. It establishes relationships between entities to ensure data integrity and provides a solid foundation for the Outcome-Based Education Management System.

7 Achievements

- Successfully developed a desktop application to automate the assessment and evaluation process.
- Implemented a user-friendly interface for easy data management by teachers.
- Ensured data integrity by applying referential integrity constraints in the database schema.
- Generated various reports including CLO wise class result, assessment wise class result, rubric wise class result, and attendance count per month.

8 Challenges

- Designing a robust database schema to accommodate complex relationships between entities.
- Implementing data validation and error handling to prevent data inconsistencies.
- Integrating report generation functionality to provide actionable insights for teachers.
- Ensuring scalability and maintainability of the application to accommodate future enhancements and changes.

9 Conclusion

The Outcome-Based Education Management System is a comprehensive solution designed to meet the assessment and evaluation needs of the Department of Computer Science at UET Lahore. By automating various tasks and providing insightful reports, the system aims to improve the overall efficiency and effectiveness of the evaluation process. With continued development and refinement, it has the potential to become a valuable tool for educators in managing student learning outcomes and enhancing the quality of education delivery.